

Probabilistic Palmprint Embeddings with Auxiliary Reconstruction for Open-Set Identification

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Abstract

Open-set palmprint identification requires reliable one-to-many matching for enrolled subjects and rejection of unknown subjects. We represent each palmprint ROI by a diagonal Gaussian whose mean carries identity information and whose variance describes input-dependent dispersion. A separate spatial pathway supplies U-Net skip features to a training-only reconstruction decoder, while a sampled latent code modulates selected skips through feature-wise linear modulation. Inference is deterministic and compares either the projected mean (M0) or the latent mean (M3); symmetric Kullback–Leibler divergence provides pairwise compatibility. The experiments on IITD and Tongji consistently show closed-set gains for CCNet, metric-dependent gains for ResNet18 and mixed results for PalmNet. The deterministic baseline PalmNet remains the strongest on Tongji and shows higher cross-dataset Rank-1 in both transfer directions. Thus, the whole probabilistic pipeline or a single inference space is not always better. The reported open-set EER, AUROC, OSCR and DIR are development-set metrics since the KL threshold is estimated from the same known-probe pool used for evaluation.

Keywords: Palmprint recognition, probabilistic embedding, open-set identification, symmetric KL divergence, FiLM, auxiliary reconstruction

1 Introduction

Palmprint recognition uses principal lines, wrinkles, ridges, and texture over a relatively large biometric surface. Deep models have improved feature extraction, but touchless acquisition is still sensitive to illumination, pose, blur, ROI localization, sensor response, and non-rigid deformation [1–4]. These factors are especially consequential in open-set identification, where a system must rank an enrolled identity when a match exists and reject a probe whose identity is absent from the gallery.

A deterministic embedding maps each observation to one point and does not explicitly encode reliability. Probabilistic Face Embeddings and Data Uncertainty Learning instead represent an image by a Gaussian mean and sample-dependent variance [5, 6]. This representation permits pairwise distribution comparison, which can incorporate uncertainty from both probe and gallery rather than reducing reliability to a probe-only scalar.

We study this formulation for palmprints using a two-pathway architecture. PalmEncoder predicts a mean and diagonal log-variance, while a separate U-Net encoder supplies spatial descriptors to a training-only decoder. The latent sample conditions selected skip through FiLM [7]; skip dropout limits deterministic bypass, a failure mode related to posterior collapse [8]. ResNet18 additionally uses a hard mean-to-sample projector switch at step 3000, motivated by curriculum learning [9]; the switch was disabled for the reported CCNet and PalmNet runs.

We deliberately restrict the contribution to (i) a Gaussian palmprint representation independent of an auxiliary spatial decoder, (ii) a training design that combines bounded log-variance, FiLM, skip dropout, MSE reconstruction, and backbone-specific loss schedules, (iii) deterministic projected-mean and latent-mean inference with symmetric-KL compatibility, and (iv) matched closed-set comparisons with deterministic ResNet18, CCNet, and PalmNet baselines on IITD and Tongji. Controlled ablations are not yet available, so we postpone causal claims at the component level.

2 Related Work

PalmNet combines Gabor processing, PCA, and convolutional learning [3], whereas CCNet uses spatial and channel competition over multi-order texture features [2]. C-LMCL and W2ML target open-set generalization through margin-based and meta-metric learning [10, 11]. More recent work tackles large-scale pattern variance, adversarial augmentation, and cross-domain alignment through RegPalm, unified adversarial augmentation, and PalmBridge [12–14]. Their protocols differ in identity partitions, gallery construction, and operating points, so published numbers are not combined with ours in a single ranking table.

Probabilistic embeddings model per-image ambiguity and enable distribution-aware matching [5, 6, 15]. Gallery-aware uncertainty further shows that recognition ambiguity depends on both a probe and its gallery neighborhood [16]. We therefore distinguish a scalar mean-log-variance diagnostic from pairwise symmetric KL. Open-set evaluation is reported with EER/AUROC diagnostics, OSCR, and DIR at fixed FPIR [17–19].

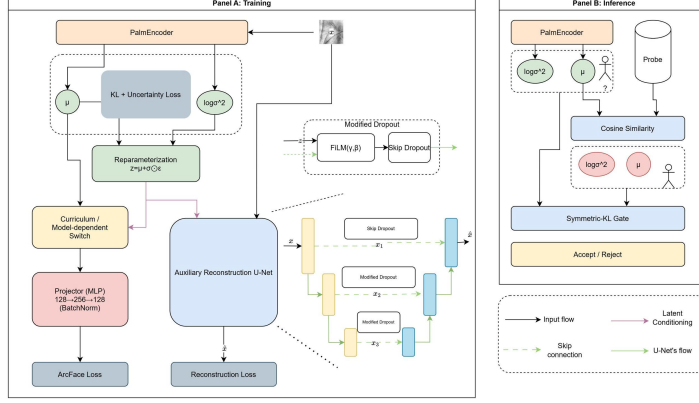


Fig. 1: Training and inference flow. PalmEncoder predicts a Gaussian representation; the spatial decoder is used only during training. M0 matches projected means, M3 matches latent means, and the compatibility check is computed from the Gaussian parameters.

Probabilistic UNet combines spatial skips with a conditional latent pathway [20]. FiLM supplies lightweight affine conditioning [7], and palmprint synthesis research stresses crease geometry and non-rigid variation [21]. In our model, the decoder is discarded at inference and is not treated as a generative identity model.

3 Method

3.1 Architecture and Auxiliary Reconstruction

For normalized ROI $\mathbf{x}_i \in [-1, 1]^{C \times H \times W}$, a U-Net encoder produces skip features $\mathbf{x}_i^{(1)}, \mathbf{x}_i^{(2)}, \mathbf{x}_i^{(3)}$. PalmEncoder predicts $\boldsymbol{\mu}_i \in \mathbb{R}^d$ and raw log-variance $\boldsymbol{\ell}_i \in \mathbb{R}^d$:

$$\begin{aligned} \tilde{\ell}_i &= \text{clamp}(\ell_i, -20, 2), & \sigma_i &= \exp\left(\frac{1}{2}\tilde{\ell}_i\right), \\ q_\theta(\mathbf{z}_i | \mathbf{x}_i) &= \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}_i^2)), & \mathbf{z}_i &= \boldsymbol{\mu}_i + \boldsymbol{\sigma}_i \odot \boldsymbol{\epsilon}_i, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \end{aligned} \quad (1)$$

The clamp limits exponentiation and extreme KL values. For $l \in \{2, 3\}$, linear layers map $\mathbf{z}_i$ to FiLM parameters:

$$\tilde{\mathbf{x}}_i^{(l)} = [1 + \gamma_l(\mathbf{z}_i)] \odot \mathbf{x}_i^{(l)} + \beta_l(\mathbf{z}_i). \quad (2)$$

The high-resolution skip $\mathbf{x}^{(1)}$ is not modulated. During training, Dropout2d with probability 0.8 removes complete skip channels. ConvTranspose2d UpBlocks and a final tanh produce $\hat{\mathbf{x}}_i \in [-1, 1]$.

3.2 Objectives and Inference

All probabilistic models use a $128 \rightarrow 256 \rightarrow 128$ projector with BatchNorm. For ResNet18 only, the projector input switches from $\boldsymbol{\mu}_i$ to $\mathbf{z}_i$ at step 3000; CCNet and

PalmNet use the configuration with this switch disabled. ArcFace uses margin $m = 0.5$ and scale $s = 30$ [22]. The remaining objectives are

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= \frac{1}{N} \sum_i D_{\text{KL}}(q_\theta(\mathbf{z}_i | \mathbf{x}_i) \| \mathcal{N}(\mathbf{0}, \mathbf{I})), \\ \mathcal{L}_{\text{rec}} &= \frac{1}{NCHW} \sum_i \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2, \\ \mathcal{L}_{\text{unc}} &= \text{ReLU}(-4 - \bar{\ell}) + \text{ReLU}(\bar{\ell} - 2), \quad \bar{\ell} = \frac{1}{Nd} \sum_{i,k} \ell_{i,k},\end{aligned}\tag{3}$$

with

$$\mathcal{L} = 0.9\mathcal{L}_{\text{id}} + 0.1\mathcal{L}_{\text{unc}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}}.\tag{4}$$

The KL and reconstruction coefficients are increased in steps. ResNet18 uses $0 \rightarrow 0.15$ and $0 \rightarrow 0.5$, CCNet uses $5 \times 10^{-6} \rightarrow 0.01$ and $0 \rightarrow 1$, and PalmNet uses $0 \rightarrow 0.005$ and $0 \rightarrow 1$. The uncertainty term constrains the batch-average log-variance; it is not a direct calibration objective.

The deterministic baseline contains a PalmEncoder and only a μ -head. It has no variance head, projector, sampling with randomness, U-Net, FiLM, KL loss, or reconstruction loss; ArcFace is applied directly to μ . At inference, M0 uses $\mathbf{f}_i = P_\psi(\mu_i)$ with BatchNorm in evaluation mode, whereas M3 uses $\mathbf{f}_i = \mu_i$. Cosine similarity is $S(p, g) = \frac{\mathbf{f}_p^\top \mathbf{f}_g}{\|\mathbf{f}_p\|_2 \|\mathbf{f}_g\|_2}$. Pairwise compatibility is

$$\begin{aligned}D_{\text{SKL}}(p, g) &= \frac{1}{2} [D_{\text{KL}}(q_p \| q_g) + D_{\text{KL}}(q_g \| q_p)], \\ D_{\text{KL}}(q_p \| q_g) &= \frac{1}{2} \sum_k \left[\frac{\sigma_{p,k}^2 + (\mu_{p,k} - \mu_{g,k})^2}{\sigma_{g,k}^2} - 1 + \log \frac{\sigma_{g,k}^2}{\sigma_{p,k}^2} \right].\end{aligned}\tag{5}$$

The highest-cosine gallery candidate is retained when its symmetric KL does not exceed τ_{KL} .

4 Experiments

4.1 Data and Training Configuration

Tongji contains 12,000 contactless images from 600 palms of 300 subjects in two sessions [23]. The supplied Tongji subject-level split uses 200 subjects for training, 50 enrolled subjects, and 50 strangers. IITD contains 230 subjects [24]. The supplied IITD subject-level split uses 160 subjects for training, 40 enrolled subjects, and 30 strangers. The three groups are identity-disjoint and are generated with split seed 42. For each enrolled subject, 50% of the images are from the gallery, and the remainder are known probes. Tongji inputs are RGB 128×128 and IITD inputs are RGB 150×150 . Augmentation uses grayscale probability 0.5, rotation $\pm 15^\circ$, brightness/contrast jitter 0.2, translation up to 0.1, and scale in $[0.9, 1.1]$.

Table 1: Training configuration. KL and reconstruction weights are linearly annealed between the reported endpoints.

Backbone	λ_{KL}	λ_{rec}	Projector switch	LR	Batch	Epochs
ResNet18	$0 \rightarrow 0.15$	$0 \rightarrow 0.5$	hard, step 3000	10^{-3}	32	100
CCNet	$5 \times 10^{-6} \rightarrow 0.01$	$0 \rightarrow 1$	off	10^{-3}	32	100
PalmNet	$0 \rightarrow 0.005$	$0 \rightarrow 1$	off	10^{-3}	32	100

Results are mean $\pm$ sample standard deviation over three seeded runs (0, 1, 2). The available logs do not establish that every seed corresponds to an independently initialized training run, so no significance test is claimed. All backbones use AdamW, batch size 32, 100 epochs, initial learning rate 10^{-3} , weight decay 5×10^{-4} , and CosineAnnealingLR. The last checkpoint is evaluated; early stopping is not used. Precision follows the PyTorch Lightning default. Runs took approximately one hour on heterogeneous machines, so hardware-normalized timing is unavailable.

4.2 Metrics and Result Validity

Closed Rank-1 is the fraction of known probes whose highest-scoring gallery identity is correct. Closed EER compares each known probe’s score to its true gallery identity with its scores to nonmatching gallery identities. For the probabilistic model, τ_{KL} is selected at the EER among genuine and known-impostor symmetric-KL distributions, and then pairs above the threshold are assigned a cosine score of -1 . O-EER and O-AUROC compare gated true-identity scores for known probes with gated maximum-gallery scores for strangers. OSCR integrates correct known classification against a stranger false-positive rate, and DIR@1% reports correct accepted known probes at an FPIR target of 1%.

The same known-probe pool is used to estimate τ_{KL} and to compute the reported open-set quantities. These values are therefore development-set diagnostics, not independently calibrated test results. Cross-dataset thresholds are also recalculated on the target known-probe pool, so those experiments are target-calibrated diagnostics rather than source-calibrated deployment.

The three deterministic baseline CSVs contain a field named **Open Rank1**, but it is not reported as standard Open Rank-1. The released evaluator applies pairwise KL masking before an identity **argmax** and does not enforce an explicit probe-level acceptance indicator. This quantity can therefore exceed closed Rank-1 and can assign a class when all candidate scores equal -1 . In addition, the deterministic CSVs contain nonzero KL/uncertainty diagnostics, although the stated baseline has no variance head. We retain only architecturally valid baseline closed metrics. DIR@0.1% is also omitted because the aggregate files do not provide trial counts sufficient to verify that operating point.

4.3 Results

The closed-set results demonstrate a considerable dependence on the backbone architecture. CCNet exhibits consistent improvement: M3 increases in-domain Rank-1 from

Table 2: In-domain results (% , mean $\pm$ sample standard deviation). Baseline open-set cells are unavailable under a verified cosine-only rejection protocol. Bold denotes the best available value in each backbone–dataset block.

Backbone	Dataset	Mode	Closed R1 $\uparrow$	C-EER $\downarrow$	O-EER $\downarrow$	AUROC $\uparrow$	OSCR $\uparrow$
ResNet18	IITD	Base	98.89 $\pm$ 0.40	1.75 $\pm$ 0.76	–	–	–
ResNet18	IITD	M0	97.51 $\pm$ 2.43	1.02 $\pm$ 0.53	2.69 $\pm$ 0.60	99.33 $\pm$ 0.33	97.61 $\pm$ 2.06
ResNet18	IITD	M3	97.79 $\pm$ 1.40	1.01 $\pm$ 0.49	2.61 $\pm$ 0.94	99.49 $\pm$ 0.17	98.18 $\pm$ 1.32
ResNet18	Tongji	Base	79.93 $\pm$ 2.78	6.47 $\pm$ 0.47	–	–	–
ResNet18	Tongji	M0	89.00 $\pm$ 1.18	4.69 $\pm$ 0.49	17.10 $\pm$ 1.22	91.67 $\pm$ 1.01	85.44 $\pm$ 1.16
ResNet18	Tongji	M3	89.40 $\pm$ 1.50	4.07 $\pm$ 0.61	15.45 $\pm$ 0.56	92.74 $\pm$ 0.82	86.69 $\pm$ 1.13
CCNet	IITD	Base	88.09 $\pm$ 1.40	17.71 $\pm$ 3.22	–	–	–
CCNet	IITD	M0	97.79 $\pm$ 2.06	1.39 $\pm$ 1.42	3.36 $\pm$ 0.65	99.17 $\pm$ 0.56	97.57 $\pm$ 2.01
CCNet	IITD	M3	99.45 $\pm$ 0.78	0.73 $\pm$ 0.52	1.70 $\pm$ 1.85	99.52 $\pm$ 0.50	99.28 $\pm$ 0.84
CCNet	Tongji	Base	61.73 $\pm$ 1.36	10.41 $\pm$ 0.84	–	–	–
CCNet	Tongji	M0	91.47 $\pm$ 1.09	3.12 $\pm$ 0.24	12.08 $\pm$ 2.04	94.80 $\pm$ 1.22	89.91 $\pm$ 1.60
CCNet	Tongji	M3	96.27 $\pm$ 1.20	2.12 $\pm$ 0.80	6.62 $\pm$ 0.70	97.59 $\pm$ 0.69	95.61 $\pm$ 0.97
PalmNet	IITD	Base	70.76 $\pm$ 3.52	9.16 $\pm$ 0.48	–	–	–
PalmNet	IITD	M0	79.21 $\pm$ 4.39	8.91 $\pm$ 2.16	28.02 $\pm$ 2.90	76.47 $\pm$ 2.93	70.79 $\pm$ 4.68
PalmNet	IITD	M3	53.66 $\pm$ 3.88	14.00 $\pm$ 2.35	45.15 $\pm$ 5.72	54.57 $\pm$ 7.03	43.97 $\pm$ 5.08
PalmNet	Tongji	Base	88.93 $\pm$ 0.25	4.36 $\pm$ 0.48	–	–	–
PalmNet	Tongji	M0	73.93 $\pm$ 2.00	8.78 $\pm$ 0.92	31.72 $\pm$ 1.63	73.75 $\pm$ 1.42	67.04 $\pm$ 1.91
PalmNet	Tongji	M3	73.07 $\pm$ 1.18	7.49 $\pm$ 0.66	33.77 $\pm$ 1.75	70.60 $\pm$ 2.14	63.21 $\pm$ 0.77

Table 3: Probabilistic in-domain diagnostics (%). KL EER and scalar EER are shown descriptively because their trial construction is not confirmed to be identical.

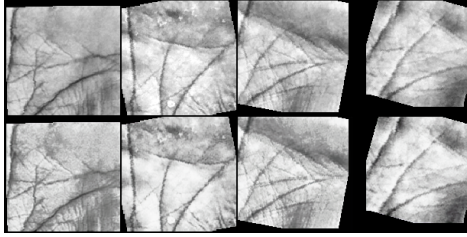
Backbone	Dataset	Mode	KL EER $\downarrow$	Scalar EER $\downarrow$	DIR@1% $\uparrow$
ResNet18	IITD	M0	1.04 $\pm$ 0.48	42.81 $\pm$ 2.75	95.29 $\pm$ 1.03
ResNet18	IITD	M3	1.04 $\pm$ 0.48	42.81 $\pm$ 2.75	95.85 $\pm$ 0.67
ResNet18	Tongji	M0	5.05 $\pm$ 0.54	48.90 $\pm$ 1.66	58.20 $\pm$ 4.54
ResNet18	Tongji	M3	5.05 $\pm$ 0.54	48.90 $\pm$ 1.66	63.00 $\pm$ 5.62
CCNet	IITD	M0	2.56 $\pm$ 1.25	55.63 $\pm$ 1.06	89.74 $\pm$ 7.60
CCNet	IITD	M3	2.56 $\pm$ 1.25	55.63 $\pm$ 1.06	98.34 $\pm$ 1.35
CCNet	Tongji	M0	4.13 $\pm$ 0.78	44.07 $\pm$ 1.21	72.07 $\pm$ 1.48
CCNet	Tongji	M3	4.13 $\pm$ 0.78	44.07 $\pm$ 1.21	86.93 $\pm$ 3.85
PalmNet	IITD	M0	19.38 $\pm$ 1.78	51.72 $\pm$ 2.42	43.33 $\pm$ 12.51
PalmNet	IITD	M3	19.38 $\pm$ 1.78	51.72 $\pm$ 2.42	14.58 $\pm$ 4.49
PalmNet	Tongji	M0	15.15 $\pm$ 0.85	46.70 $\pm$ 2.90	31.53 $\pm$ 7.57
PalmNet	Tongji	M3	15.15 $\pm$ 0.85	46.70 $\pm$ 2.90	25.93 $\pm$ 5.05

88.09% to 99.45% on IITD and from 61.73% to 96.27% on Tongji, while also reducing EER. ResNet18 shows gains on Tongji and in both transfer directions; however, its deterministic IITD baseline maintains the highest in-domain Rank-1 (98.89%), with M0 and M3 instead improving IITD EER to approximately 1.0%. PalmNet yields mixed results. M0 increases IITD Rank-1 from 70.76% to 79.21%, but the deterministic baseline outperforms on Tongji (88.93% Rank-1 and 4.36% EER). In cross-dataset evaluation, the PalmNet baseline achieves the highest Rank-1 in both directions, and the small EER reductions observed with M3 fall within the reported run variability.

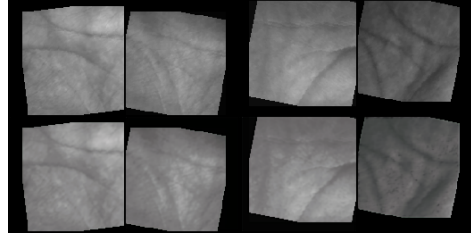
Within the probabilistic model, M3 generally yields the strongest performance for both ResNet18 and CCNet. PalmNet demonstrates a preference for M0 in-domain, although M3 performs better on cross-dataset open-set diagnostics. Scalar EER remains close to chance (42.81–55.63%), while KL EER is substantially lower (1.04–19.38%). As the corresponding trial populations have not been confirmed as identical, this difference supports only a descriptive observation that pairwise KL is

Table 4: Cross-dataset results (%). Models are not fine-tuned on the target dataset; open-set thresholds are recalculated on the target known-probe pool.

Backbone	Train	Test	Mode	Closed R1 $\uparrow$	C-EER $\downarrow$	O-EER $\downarrow$	AUROC $\uparrow$	OSCR $\uparrow$
ResNet18	IITD	Tongji	Base	50.40 $\pm$ 4.87	26.84 $\pm$ 1.73	—	—	—
ResNet18	IITD	Tongji	M0	71.47 $\pm$ 1.23	11.27 $\pm$ 1.46	36.67 $\pm$ 2.87	68.74 $\pm$ 4.15	57.99 $\pm$ 3.05
ResNet18	IITD	Tongji	M3	72.27 $\pm$ 1.31	12.24 $\pm$ 0.93	35.73 $\pm$ 3.61	69.01 $\pm$ 4.71	58.58 $\pm$ 4.39
ResNet18	Tongji	IITD	Base	74.26 $\pm$ 5.28	15.59 $\pm$ 1.37	—	—	—
ResNet18	Tongji	IITD	M0	81.72 $\pm$ 1.72	10.66 $\pm$ 0.92	24.63 $\pm$ 2.52	79.38 $\pm$ 3.05	74.30 $\pm$ 2.23
ResNet18	Tongji	IITD	M3	84.22 $\pm$ 2.38	10.13 $\pm$ 1.51	23.52 $\pm$ 3.59	80.91 $\pm$ 3.21	76.38 $\pm$ 2.65
CCNet	IITD	Tongji	Base	53.13 $\pm$ 1.31	20.93 $\pm$ 1.60	—	—	—
CCNet	IITD	Tongji	M0	79.00 $\pm$ 1.82	10.67 $\pm$ 0.26	32.17 $\pm$ 1.47	75.44 $\pm$ 1.56	67.43 $\pm$ 0.79
CCNet	IITD	Tongji	M3	90.47 $\pm$ 2.13	9.65 $\pm$ 0.90	26.88 $\pm$ 2.59	81.64 $\pm$ 2.26	77.69 $\pm$ 2.28
CCNet	Tongji	IITD	Base	71.76 $\pm$ 4.36	13.61 $\pm$ 2.34	—	—	—
CCNet	Tongji	IITD	M0	91.70 $\pm$ 3.74	4.55 $\pm$ 2.13	12.57 $\pm$ 2.25	93.22 $\pm$ 2.71	90.54 $\pm$ 3.56
CCNet	Tongji	IITD	M3	98.07 $\pm$ 1.56	1.78 $\pm$ 0.64	7.44 $\pm$ 3.99	96.68 $\pm$ 1.69	96.05 $\pm$ 2.35
PalmNet	IITD	Tongji	Base	43.33 $\pm$ 3.84	22.05 $\pm$ 1.87	—	—	—
PalmNet	IITD	Tongji	M0	31.27 $\pm$ 5.57	22.80 $\pm$ 1.69	63.92 $\pm$ 1.65	31.51 $\pm$ 3.41	19.63 $\pm$ 4.56
PalmNet	IITD	Tongji	M3	33.80 $\pm$ 7.19	21.98 $\pm$ 1.68	62.60 $\pm$ 3.36	33.38 $\pm$ 4.27	22.87 $\pm$ 5.04
PalmNet	Tongji	IITD	Base	35.51 $\pm$ 1.89	28.40 $\pm$ 0.96	—	—	—
PalmNet	Tongji	IITD	M0	13.99 $\pm$ 1.35	37.47 $\pm$ 0.49	70.74 $\pm$ 1.12	23.01 $\pm$ 1.23	7.95 $\pm$ 0.94
PalmNet	Tongji	IITD	M3	26.79 $\pm$ 3.09	28.20 $\pm$ 0.94	64.48 $\pm$ 1.55	31.32 $\pm$ 0.81	16.44 $\pm$ 2.80



(a) IITD



(b) Tongji

Fig. 2: CCNet reconstruction examples: inputs are shown above reconstructions. Major crease layout and pose remain visible, while local contrast is attenuated. No quantitative reconstruction claim is made.

more separable with the current evaluator; it does not provide evidence of calibrated uncertainty.

5 Discussion

The CCNet results are the most convincing evidence: both closed-set metrics improve significantly in-domain and cross-dataset. ResNet18 is less useful. Improvements in EER and transfer metrics are observed but the deterministic IITD baseline still has a higher Rank-1. PalmNet is a counterexample to a universal advantage because M0 improves IITD performance, however the deterministic model outperforms on Tongji and both cross-dataset Rank-1 comparisons. These results support conclusions specific to the model and dataset rather than a general claim of superiority for probabilistic training or a specific inference space.

The present experiment does not disentangle the specific contributions of the variance head, KL loss, decoder, FiLM, skip dropout or stochastic latent input. Key control

conditions would include deterministic-plus-projector, variance without decoder, omission of $\mathcal{L}_{\text{KL}}$ or $\mathcal{L}_{\text{rec}}$, constant or shuffled variance, omission of FiLM, omission of skip dropout, and zero or shuffled $\mathbf{z}$. Reconstructions with nonconstant outputs are shown in Figure 2, but it does not show that the decoder uses $\mathbf{z}$ instead of skip features.

The primary limitation of the current evaluation is still independent calibration. A better protocol would be to reserve calibration to known and stranger probes, set all thresholds prior to evaluation and measure performance on disjoint test probes. It should also report absolute numbers of false accepts and per-probe trial counts and present uncertainty or risk-coverage analysis. The aggregate files as they currently exist do not support claims of statistical significance, feature-hash verification or a reliable DIR@0.1% operating point.

Palmprint images and embeddings are persistent biometric data. Raw ROIs, means, variances and gallery templates should be treated as sensitive information. This study does not include template protection, inversion, membership inference, presentation attacks or demographic effects.

6 Conclusion

We propose Gaussian palmprint embeddings, a training-only MSE decoder, FiLM-modulated skip connections, projected-mean and latent-mean inference, and symmetric-KL compatibility. CCNet consistently improves, ResNet18 improves depending on the metric, and PalmNet has mixed results. The outcomes support a backbone- and dataset-dependent evaluation, not a general claim for probabilistic training or a specific inference space. We need to do independent calibrations and controlled ablation studies before we can say anything about gains from uncertainty modeling or reconstruction.

Declarations

Data availability. IITD and Tongji are publicly available datasets, no new biometric data was collected. Source code can be found at [the project repository](#). The repository must be version tagged and in sync with the final protocol before submission.

Ethics and privacy. Palmprint images and embeddings shall be considered sensitive biometric data and shall be used in accordance with the dataset and institutional requirements. **Conflicts of Interest.** None declared.

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